

5.4 Nuclear decay

This topic covers radioactive decay.

This topic may be taught using applications that relate to, for example, medical physics.

Concept approach

Students will be assessed on their ability to:	Suggested experiments
114 show an awareness of the existence and origin of background radiation, past and present	Measure background count rate
115 investigate and recognise nuclear radiations (alpha, beta and gamma) from their penetrating power and ionising ability	Investigate the absorption of radiation by paper, aluminium and lead (radiation penetration simulation software is a viable alternative)
116 describe the spontaneous and random nature of nuclear decay	
117 determine the half lives of radioactive isotopes graphically and recognise and use the expressions for radioactive decay: $dN/dt = -\lambda N$, $\lambda = \ln 2/t_{1/2}$ and $N = N_0 e^{-\lambda t}$	Measure the activity of a radioactive source Simulation of radioactive decay using, for example, dice
118 discuss the applications of radioactive materials, including ethical and environmental issues	